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PI Name:	Fornace, Albert M.D.		
Project Title:	NSCOR: Space Radiation and Gastrointestinal Cancer: A Comprehensive Strategy for Risk Assessment and Model Development		
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Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
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Space Biology Special Category:	None		
PI Email:	af294@georgetown.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	202 687-7843
Organization Name:	Georgetown University		
PI Address 1:	Dept. of Oncology, Lombardi Comprehensive Cancer Center		
PI Address 2:	Research Building, Room E504, 3970 Reservoir Rd., NW		
PI Web Page:			
City:	Washington	**State:**	DC
Zip Code:	20007-2126	**Congressional District:**	1
Comments:	http://www9.georgetown.edu/		
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Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
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Key Personnel Changes/Previous PI:	July 2022 report: Shubhankar Suman added as Co-Investigator in the year-6 of this NASA Specialized Center of Research (NSCOR).		
COI Name (Institution):	Brenner, David Ph.D. (Columbia University) Meltzer, Paul M.D., Ph.D. (National Cancer Institute, NIH) Shay, Jerry Ph.D. (University of Texas Southwestern Medical Center at Dallas) Suman, Shubhankar Ph.D. (Georgetown University)		
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Performance Goal Text:	
Task Description:	Considering the high spontaneous incidence of gastrointestinal (GI) and lung cancer, and even a modest increase by space radiation exposure could have a significant effect on astronauts' health risk estimates during and after long-duration manned space flights. However, there is substantial uncertainty for GI and lung cancer risk estimation from space radiation due to the lack of in vivo human data. Our overall goal is to specifically investigate cellular and molecular hypothesis-driven mechanisms in relevant mouse models that will contribute to carcinogenic risk estimates of GI and lung cancer after exposure to space radiation beams prioritized by NASA's human research program (HRP). Specifically, we seek to determine incidence, tumor frequency, and grade as well as identify molecular perturbations in the target cells associated GI and lung tumorigenesis through monitoring of changes in the tumor number, histology, and gene/protein expression of the proposed model system after exposure to space-radiation type beams. The overall hypothesis is that neutrons and mixed beam GCR type radiation are more carcinogenic due to important qualitative differences from ? rays where human exposure data are available. Molecular events triggered by space radiation will be investigated in-depth and the mechanistic basis for persistent aberrant cell-signaling after space radiation will be dissected. Both quantitative and qualitative murine tumor data will contribute to the modeling of human risks for GI and lung cancer by space travel. We plan to test our hypothesis by pursuing the following specific aims: Aim 1. Quantitatively assess GI tumorigenesis in mouse models of GI cancer and collect samples for qualitative analysis. Aim 2. Dissection of the signaling events and consequences in GI cells of the persistent effects of space radiation. Aim 3. Quantitatively assess lung tumorigenesis and test potential countermeasures in lung cancer susceptible mouse model. Aim 4. Development of mathematical modeling for cancer risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gastrointestinal tumors are frequent in the U.S, and the American Cancer Society estimates that there will be 147,950 new cases of colorectal cancer (CRC) with 53,200 persons predicted to die of the disease in 2020. Overall, the lifetime risk of developing colorectal cancer is about 1 in 23 (4.4%) for men and 1 in 25 (4.1%) for women. Considering the high frequency of colorectal tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. In addition to risk estimation, studies on the persistence of oxidative and inflammatory stress after HZE radiation and its role in driving gastrointestinal tumorigenesis may provide insight into mutagenic processes affecting genome integrity and carcinogenesis. The significance and deliverable of this project are to improve the estimates of GI cancer risk in astronauts and to identify and identify plausible targets for the future development of mitigation strategies.
	Pointwise progress made for each aim/project is summarized below. GI-tumorigenesis study (Project 1, Georgetown University): In the last six years of NASA Specialized Center of Research (NSCOR) support, we have successfully applied mouse model approaches to acquire quantitative and qualitative gastrointestinal (GI) tumorigenesis data to address prevailing uncertainties in risk modeling. Our overall goal is to determine the relative biological effectiveness (RBE) of various parameters for space radiation compared to terrestrial radiation. The risk estimates can then be extrapolated to space radiation using an RBE scaling factor. Our strategy is to: i] Quantitatively analyze tumor incidence and grade following space type radiation exposure. ii] Assess potential dose rate and gender variation. iii] Delineate cancer progression-specific biological signatures associated with space radiation exposure. Since low-linear energy transfer (LET) gamma radiation risk estimates are available for human populations from studies in atom bomb cohorts and the like, we will then be able to combine our data to model the relative risk for space radiation. Pointwise details of progress made for GI tumorigenesis study are provided below: 1. We participated in the NASA Space Radiation Laboratory / NSRL-21B (summer) and NSRL-22B (summer) beam campaigns. During these beam runs, we conducted a pilot GI-tumorigenesis experiment with a limited number of male and female Apc1638N/+ mice exposed to acute (25 to 75 cGy) and chronic full-spectrum galactic cosmic radiation (GCR) beams. 2. Sequentially delivered proton (H) to Helium (He) to Oxygen (O) to Silicon (Si) beams that were designed to simulate simplified-mixed-field GCR. A comparison of tumor data from simplified-mixed-field GCR and equivalent doses of heavy ions revealed an association between higher GI-tumorigenesis where the dose received from heavy ions contributed to >95% of the total GI-tumorigenic effect observed after mixed-field GCR. This study provides the first experimental evidence that cancer risk after GCR exposure could largely depend on doses received from constituent heavy-ions. 3. Aspirin, a well-known inhibitor of the cyclooxygenase-2/prostaglandin E2 (COX/PGE2) pathway, was evaluated as a potential countermeasure against 28Si-induced PGE2 and GI-tumorigenesis in Apc1638N/+. Aspirin led to a significant reduction in PGE2 in a dose-dependent manner but did not reduce 28Si-induced GI tumorigenesis even at the highest (300 mg/day) dose. In summary, this study suggests that aspirin has no preventive effect against high-LET radiation-induced GI tumorigenesis in Apc1638N/+ mice. 4. Metformin is known to suppress insulin-like growth factor 1 (IGF1) signaling, via activation of AMP-activated protein kinase (AMPK). Here we tested metformin for its potential GI-cancer preventive effects against heavy-ion radiation-induced GI-tumorigenesis. Male Apc1638N/+ mice were continuously fed either standard or metformin supplemented diet (0.1% W/W) starting at the age of 4 weeks, and one month later mice were irradiated with 10 cGy of 28Si at the NSRL. At 150-day post-exposure intestinal tumor number, cluster, size, and tumor grade were assessed. The systemic impact of metformin on serum IGF1 was analyzed using an antibody array. We have reported that long-term continuous intake of metformin confers significant protection against heavy-ion radiation-induced GI-tumorigenesis in Apc1638N/+ mice. Here we report that metformin has a preventive effect against heavy ion-induced serum level IGF1, a pro-carcinogenic growth factor with an established role in GI-cancer promotion. Dissection of the signaling events and consequences in GI cells of the persistent effects of space radiation (Project 2, Georgetown University): Proliferative gastrointestinal (GI) tissue is radiation-sensitive; and heavy-ion space radiation, with its high-linear energy transfer (high-LET) and higher damaging potential than low-LET gamma-rays, is predicted to compromise astronauts' GI function. However, much uncertainty remains in our understanding of how heavy ions affect coordinated epithelial cell migration and extrusion, which are essential for GI homeostasis. The overall goal of this project is to understand the long-term effects of space radiation on normal gastrointestinal (GI) cells, including stem cells. Coordinated epithelial cell migration is key to maintaining functional integrity and preventing pathological processes in gastrointestinal (GI)

Task Progress:	tissue and is essential for astronauts' health and space mission success. Pointwise details of progress made for persistent effects of space radiation on GI-tissues are provided below: 1. Heavy ion radiation exposure induces senescence, and triggers accelerated senescence-associated secretory phenotype (SASP) in mouse intestinal Lgr5+ stem cells. Higher levels of senescence and SASP were observed at 1 year compared to the 60d timepoint. 2. ß-gal, lysozyme, IL1R, or IL1ß showed the highest staining in 28Si irradiated compared to control intestinal stem cells (ISCs) samples. Phospho-Histone H3 co-staining showed a mutually exclusive staining pattern with a higher level of proliferating cells in 28Si compared to control or gamma-rays irradiated groups. g-H2AX staining showed greater foci numbers in 28Si relative to gamma-rays or the control group. 3. The higher expression of secretory cytokines IL8 and IL15 was observed in 28Si irradiated ISCs relative to gamma-rays or the control group at 60d and 1y time points. Enhanced expression of Dickkopf-related protein (DKK2) and activation of Nuclear factor kappa B and p38 mitogen-activated protein kinase (MAPK) were visualized in ß-gal positive cells of 28Si irradiated ISCs relative to gamma-rays or control group at these time points. 4. Senescent cells can be removed intrinsically by immune cells; however, we know very little about the effects of heavy ion radiation on immune cells. Flow cytometry analysis showed a significant decrease in NKp46 (natural killer/NK cell marker) expression in 28Si relative to the gamma-ray or control group. Our results demonstrate that 28Si exposure decreased NK cell infiltration into the Lamina Propria, and inefficient senescent surveillance may contribute to the persistent accumulations of senescent cells, and increased SASP signaling. Lung tumorigenesis and countermeasure studies (Project 3, University of Texas Southwestern Medical Center / UTSW): This project involves lung tissue from normal wildtype (WT) mice, and a mouse model of lung cancer (LA-1). The rationale for these models is that whole body exposure to space radiation will affect organ/tissues and cells in different ways. The cells in the lung only turn over with damage, while the colon turns over in humans every 7 days irrespective of damage. Thus, to model increased risks of more lethal cancers from space radiation exposure, it is important to examine tissues with both a high rate of cellular turnover (colon, Georgetown plan) and tissues with a low rate of cell turnover (lung, UTSW plan). The inclusion of a lung model reflects the major solid tumor risk for astronauts traveling in space. Results from these studies were designed to evaluate the effects of both acute and chronic doses of the GCR Simulator on radiation-induced health risks using a radiation-induced lung cancer in male and female wildtype and LA1 mutant (a lung cancer susceptible mouse) mice on a 129 background. Each study included an acutely exposed group, a fractionated exposed group with consecutive doses of 20.8 mGy/day delivered over a four-week period, and sham controls. The simulator was baselined to 50 cGy for the first series of experiments. Acute exposures were delivered halfway through the fractionated exposure to support age-matching post irradiation. Recorded exposure times for either the acute or fractionated exposure were ~70 to 75 minutes, with the largest portion of time devoted to beam switching. During this initial experimental campaign, approximately 300 mice received daily irradiations and ~200 mice treated as sham controls. This required the additional daily loading and unloading of all control animals from the cage array over the four-week period as well. Based on camera footage from the target room, the mice adapted nicely to the exposure boxes and remained active especially with the nestlet/bedding material provided. This was the case even after being loaded into and out of the cages over the course of four weeks and alleviated concerns over the possibility of mice being shielded by cage mates while sleeping in the corners and not receiving a more uniform dose in all directions. We then initiated a series of GCRsim with acute and chronic 75cGy but the results are not complete yet. Even with 75cGy total dose, there is only 6.3cGy of HZE >3. Finally, the Brookhaven National Laboratory (BNL) team irradiated our mice with a 25cGy and 100cGy acute GCRsim during the NSRL 20B, and results are still in progress. Thus, there are experiments that need to be completed. The survival studies and overall incidence of invasive, more lethal cancers require a year post irradiation. At NSRL-22A, we irradiated an additional 100 mice with acute 75cGy GCRsim, With sufficient mice for 22B, we also irradiated mice each with acute 25, 50, and 100cGy 33 beam GCR to increase our numbers of mice from previous experiments to show reproducibility and to determine if there are sex differences in cancer initiation and progression. Cancer risk modeling (Project 4, Columbia University): A trip to Mars and back, or another lengthy deep space exploration mission, will result in exposure of astronauts to a multi-component mixture of ionizing radiations which damage cells in multiple ways. Such damage mechanisms can be roughly classified into two categories: (1) Targeted effects (TE), involving the consequences of direct traversals of cells by ionizing tracks leading to DNA double strand breaks and other lesions. (2) Non-targeted effects (NTE, or "bystander" effects) caused by release of signals from cells directly hit by the tracks and the effects of these signals on other cells. The presence of both types of effects generates complex dose response shapes for space radiations, particularly in the low dose region relevant for space missions. Here we used our mechanistically-motivated model formalism that includes both TE and NTE to analyze updated mouse tumorigenesis data from the current NSCOR. The updated data set now includes more detailed information on space-relevant doses of multiple radiation types. This new information and improved modeling approach allowed better estimation of dose response shapes and radiation effectiveness metrics in the space-relevant dose region. Specifically, we predicted the tumor yield for a Mars-mission-relevant mixture of radiation types, using the new Incremental Effect Additivity (IEA) synergy theory, which addresses many of the weaknesses of the simple effects additivity (SEA) theory that is commonly used. These improvements can provide enhanced estimation of cancer risks in astronauts.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
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